Learning the Rhythm: Why RoPE Degrades Beyond the Training Horizon, and How Learnable Frequencies Help

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Problem & Background

Problem: one model, many horizons

Finance needs forecasts at several horizons at once.

- Risk: Value-at-Risk at **1 day** and **10 days**; volatility daily $\rightarrow$ weekly $\rightarrow$ monthly (HAR).
- Energy trading: bid for **tomorrow**, hedge for **next week**.
- Usual practice: *one model per horizon* — costly to train and maintain.

Goal

Train once on a short horizon, then forecast **any** horizon:

$$\underbrace{x_{t-335}, \ldots, x_t}_{L=336 \text{ past steps}} \longmapsto \underbrace{\hat{x}_{t+1}, \ldots, \hat{x}_{t+H'}}_{\text{trained with } H=96,\ \text{asked for } H' \leq 720}$$

A **step** = one observation. On hourly prices, 96 steps = **4 days**, 720 steps = **30 days**.

The problem: silent failure beyond the training horizon

What happens. Trained for 96 steps (4 days), asked for 720 (30 days): the error rises sharply beyond the training horizon.

Why it matters.

- **Silent**: fine at the training horizon, no warning beyond it.
- **Common**: RoPE is widely used in LLMs and time-series models.
- **Costly**: the alternative is one model per horizon.

6.4×

error at 720 steps vs. a model trained for 720 (Electricity)

Our explanation (evidence on pages 12, 18–19)

RoPE's **fixed periods** do not match the **daily cycle**: each day the same hour gets a slightly different position code, so far ahead the model meets codes it never saw.

Where we started: the proposal

Proposal idea.

- HAR fixes horizons by hand (day / week / month).
- Give each head a **learnable time scale** on its rotary encoding; let the model *learn* which scales matter.

What we actually found.

- At the training horizon the scales **barely move** ($\pm 10\%$); accuracy changes little.
- Heads learn **similar** scales.
- Beyond the training horizon they make a **large** difference.

This talk

Why does RoPE degrade beyond the training horizon, how much do 4 parameters per layer help, and what does it mean for finance?

Rotary Position Embedding (RoPE)

Definition (Su et al., 2021, *RoFormer*)

For each attention head of dimension d_h (even), split into $m = d_h/2$ two-dimensional blocks. At position ℓ, RoPE applies a rotation with frequency ω_i:

$$\mathrm{RoPE}(x, \ell) = \begin{bmatrix} x_{2i}\cos(\omega_i \ell) - x_{2i+1}\sin(\omega_i \ell) \\ x_{2i}\sin(\omega_i \ell) + x_{2i+1}\cos(\omega_i \ell) \end{bmatrix}, \qquad \omega_i = \theta^{-2i/d_h}.$$

Key Property

The dot product between a query–key pair after RoPE depends only on their **relative offset**:

$$\langle \mathrm{RoPE}(q, \ell_q),\, \mathrm{RoPE}(k, \ell_k) \rangle = \langle q,\, R(\omega_i(\ell_k - \ell_q))\, k \rangle.$$

Hence attention sees **relative distances** only — what matters for time series is the lag, not the calendar index.

RoPE Mechanism — Frequency Interpretation

Expand one 2D block. Let $q^{(i)} = [q_{i,1}, q_{i,2}]^\top$, $k^{(i)} = [k_{i,1}, k_{i,2}]^\top$:

$$q^{(i)\top} R(\omega_i \Delta) k^{(i)} = (q_{i,1}k_{i,1} + q_{i,2}k_{i,2})\cos(\omega_i\Delta) + (q_{i,2}k_{i,1} - q_{i,1}k_{i,2})\sin(\omega_i\Delta).$$

Define coefficients.

$$a_i = \langle q^{(i)}, k^{(i)} \rangle, \quad b_i = q_{i,2}k_{i,1} - q_{i,1}k_{i,2}.$$

Trigonometric form

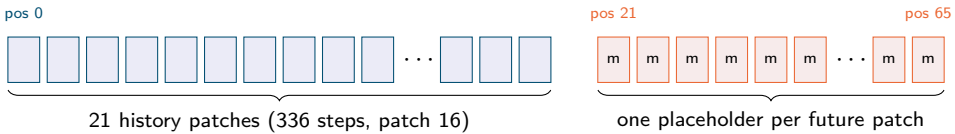
$$S(\Delta) = \sum_{i=0}^{m-1} [a_i \cos(\omega_i \Delta) + b_i \sin(\omega_i \Delta)].$$

Interpretation. Each block acts as a **sinusoidal filter**: large $\omega_i \to$ short-term/high-frequency; small $\omega_i \to$ long-term/low-frequency.

Head-wise scaling (our proposal). $\omega_i^{(h)} = \omega_i \, e^{a_h}$ was meant to let heads specialise on distinct horizons. In our experiments the learned scale instead moves the **fastest** block ($\omega_0 = 1$ rad/token, period $2\pi \approx 6.28$ tokens) close to the daily cycle.

Method

Model: placeholders for the future



- 2-layer, 4-head Transformer encoder; $d = 64$. Each history patch of 16 steps is one token.
- All placeholders are the **same** learnable vector `m`: **only RoPE tells them apart**.
- Mask: history sees history; a placeholder sees history and itself $\Rightarrow$ the forecast for step h does not depend on H, so we can ask for **any** horizon after training.
- Trained with $H = 96$ (6 placeholders); queried with $H = 720$ (45 placeholders).

Our method: learnable frequency scales

Per-head learnable scale (the proposal's idea)

$$\hat{\omega}_{\ell,h,j} = \omega_j \cdot e^{a_{\ell,h}}, \qquad a_{\ell,h} \text{ learned, initialised at } 0 \ (= \text{standard RoPE})$$

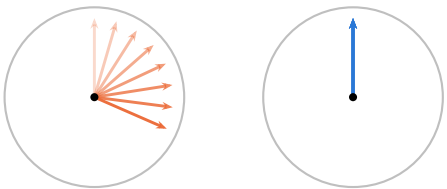
Only **4 extra parameters per layer** (one per head).

Ablation variants (to find out *why* it works):

- *Grouped*: separate scales for the fast and the slow frequency pairs.
- *Shared*: one scale for all heads of a layer.
- *Fixed*: scales set by hand and frozen (spread across heads, or aligned to the day).

Second remedy — multi-horizon training: each batch uses $H \in \{96, 192, 336, 720\}$, so one model also sees distant placeholders.

Intuition: RoPE as a clock



Standard RoPE ($\omega_0 = 1$) Aligned ($\omega_0 = 2\pi/6$)

fastest hand at midnight on days 0–7 (darker = later)

- ETTm1: 15-min data, patch 16 $\Rightarrow$ **1 day = 6 tokens**.
- Standard RoPE turns once per **6.28** tokens: it falls behind by 0.28 tokens a day.
- Inside the training window the drift is small; 7 days ahead the hand points somewhere **never seen in training**.

Alignment

If $\hat{\omega}_0 = 2\pi/T$ (T = tokens per day), position $t + T$ looks exactly like t: the far future looks like the past.

Experiments & Mechanism

Experimental setup

Data (7 datasets, every variable its own series)

- Benchmarks: Electricity, ETTh1, ETTm1, Weather.
- **Financial**: day-ahead electricity prices Nord Pool (EPF-NP) & PJM (EPF-PJM); daily exchange rates.
- Chronological splits; z-score on training rows only; RevIN for exchange rates.

Metric: MSE on the normalised scale.

Baselines & why

- *Fixed RoPE*: same model without scales $\Rightarrow$ isolates the mechanism.
- *Linear* $\mathbb{R}^{336} \to \mathbb{R}^{H}$: as strong as Transformers here (DLinear).
- *Trained directly at 720*: the natural alternative to extrapolation.

Protocol & significance

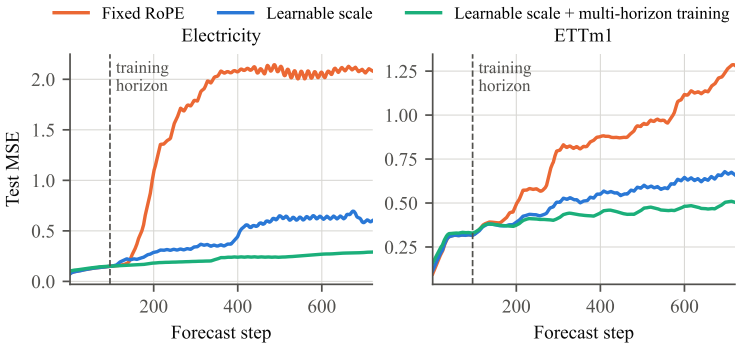
- Same backbone, paired seeds, base and checkpoint chosen on validation only.
- 3–5 seeds + **Diebold–Mariano** test (Newey–West, overlapping windows).

Result 1: at the training horizon

Test MSE, $H = 96$	Linear	Transformer, fixed RoPE	Transformer, learnable scale
Electricity	0.140	0.139	0.139
ETTh1	**0.377**	0.417	0.414
ETTm1	**0.304**	0.311	0.317
Weather	0.174	**0.153**	**0.153**
EPF-NP	0.472	0.482	**0.452**
EPF-PJM	**0.163**	0.174	0.167
Exchange	**0.089**	0.103	0.099

- The Transformer is close to published results (PatchTST/42: Electricity 0.130, Weather 0.152).
- Linear models remain strong (as reported by DLinear).
- Fixed vs. learnable differ by about one seed std or less here — the main effect lies **beyond** the horizon.

Result 2: fixed RoPE degrades beyond it



Test MSE per forecast step; trained at $H = 96$ (dashed line).

6.4×

error of fixed RoPE at 720 steps vs. a model trained at 720 (Electricity)

38–66%

lower error with learnable scales (Electricity, ETTh1, ETTm1)

6 / 6

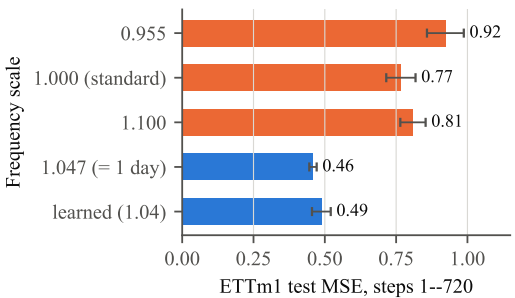
runs improve, each significant (Diebold–Mariano), on all three

Result 2: all datasets at 720 steps

	Trained at 96, asked for 720		DM	Multi-horizon	Trained at 720	
Test MSE, steps 1–720	fixed	learnable	+ / −	learnable	Transformer	Linear
Electricity	1.347±0.087	0.464±0.107	6 / 0	0.216±0.003	0.210±0.002	0.204±0.000
ETTh1	0.972±0.071	0.473±0.013	6 / 0	0.541±0.014	0.632±0.036	0.493±0.004
ETTm1	0.764±0.068	0.474±0.019	6 / 0	0.421±0.011	0.435±0.013	0.430±0.003
Weather	0.392±0.009	0.396±0.006	2 / 1	0.326±0.009	0.336±0.004	0.325±0.001
EPF-NP	1.072±0.061	0.896±0.039	5 / 0	1.123±0.008	1.196±0.156	0.863±0.003
EPF-PJM	0.443±0.036	0.273±0.034	6 / 0	0.216±0.002	0.213±0.003	0.221±0.000
Exchange	1.078±0.100	1.024±0.036	3 / 3	1.074±0.087	1.097±0.214	0.890±0.003

- Strong, significant gains on seasonal data: Electricity, ETTh1, ETTm1 and **both electricity markets**.
- No consistent gain on Weather and Exchange rates (DM results of mixed sign) — explained on the next slides.
- DM column: of 6 runs (3 seeds $\times$ 2 bases), how many are significantly better / worse ($p < 0.05$).

Why: alignment to the daily cycle

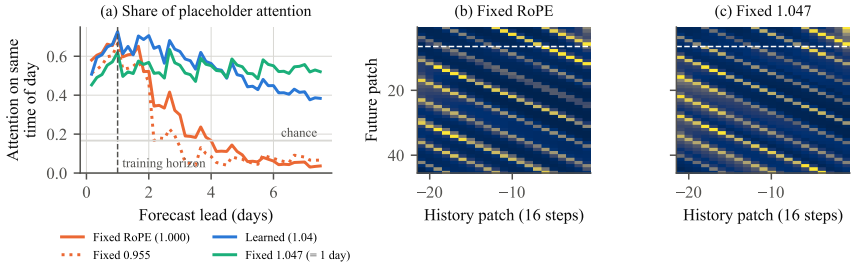


- ETTm1: 1 day = 6 tokens $\Rightarrow$ aligned scale $2\pi/6 = \mathbf{1.047}$.
- Learned scales average **1.04** — in both layers.
- **Fixed at 1.047, no learning**: matches the learned model.
- 5% too slow (0.955) or fast (1.10): degrades like standard RoPE.
- DM test 1.047 vs 0.955: $p < 10^{-15}$ in 6/6 runs.

Explanation supported by the evidence

Learning helps by moving **RoPE's fastest rotation close to the daily cycle**.

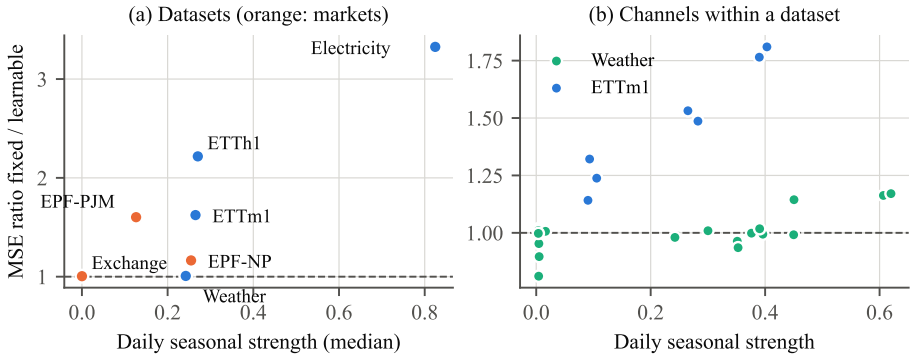
Attention: what the placeholders look at



ETTm1, trained at $H = 96$, queried at 720; last layer, mean of 3 seeds; dashed line = training horizon.

- Future patches attend to history at the **same time of day** (stripes every 6 patches): a "same time yesterday" lookup.
- **Fixed RoPE loses it** after $\sim$2 days ($0.64 \to 0.05$, below chance); **aligned 1.047 keeps it** (≈ 0.53); learned 1.04 decays slowly.

Gain grows with the daily cycle



- **Datasets**: strongest cycle (Electricity) gains most; no cycle (Exchange): no consistent effect.
- **Channels**: in Weather only the 3 solar-radiation channels gain — so Weather as a whole shows none.

Remedy 2: multi-horizon training

MSE, 720 steps	Learnable @96	Multi-H (1 model)	Trained @720
Electricity	0.464	**0.216**	0.210
ETTm1	0.474	**0.421**	0.435
Weather	0.396	**0.326**	0.336
EPF-PJM	0.273	**0.216**	0.213
ETTh1	**0.473**	0.541	0.632
EPF-NP	**0.896**	1.123	1.196

- One model trained on mixed horizons is within 3% of, or better than, the model trained at 720 — on **all 7 datasets**.
- Small cost at $H = 96$ (Electricity 0.135 vs 0.128).
- With it, fixed and learnable RoPE perform alike (distant positions are seen in training).
- When long-horizon training **overfits** (small ETTh1, noisy EPF-NP), learnable scales + extrapolation do better.

Application to Finance

Application to finance

Electricity markets (daily cycle)

Learnable scales cut 720-step error significantly:
EPF-PJM $0.44 \rightarrow \mathbf{0.27}$ (6/6 runs),
EPF-NP $1.07 \rightarrow \mathbf{0.90}$ (5/6 runs).

Exchange rates (random walk)

No cycle to align $\Rightarrow$ no consistent effect (3 better, 3 worse); linear model best; RevIN essential.

Practical guidance from our results

1. For **seasonal** financial series — intraday volume & volatility (daily U-shape), power & gas prices — use learnable or calendar-aligned RoPE before forecasting beyond the training horizon.
2. To cover several risk horizons (1- & 10-day VaR, HAR-style volatility) with **one model**, train on mixed horizons.
3. For near-random-walk prices, we saw no benefit from position encodings.

Limitations & Takeaways

Limitations & future work

What did not work / limitations

- No evidence that heads specialise to distinct time scales (the proposal's hypothesis).
- Mechanism analyses cover ETTh1, ETTm1 and Weather only.
- Grouping frequencies (the starting repository's hypothesis) gave no extra gain.
- Small model (2 layers, $d = 64$), 3–5 seeds (partly offset by DM tests).
- With point tokens instead of patches the effect is much noisier; linear models stay strong at the training horizon.

Future work

- Initialise frequencies from the data's dominant period (FFT) instead of learning them.
- Align several frequencies to several cycles (day *and* week).
- Test on larger Transformers and time-series foundation models that use RoPE.

Takeaways

1. The problem

In our experiments, RoPE forecasters trained at one horizon **degrade sharply** beyond it: up to $6.4\times$ the error of a directly trained model.

2. The likely cause

RoPE's rotation is **out of step with the daily cycle**; far-ahead positions look unlike those seen in training.

3. The remedies

4 learnable scales per layer **align** it (38–66% less error, consistent across runs); one **mixed-horizon** model comes within 3% of per-horizon models.

For finance

Helped on seasonal markets (electricity prices); no consistent effect on random-walk prices (exchange rates).

Thank you!

Questions?

Backup: What Does *Not* Matter

Frequency scales	Test MSE
ETTh1, steps 1–336, 5 seeds	
Fixed RoPE (scale 1)	0.676±0.022
Learnable, per head	0.460±0.006
Learnable, grouped	0.460±0.006
Learnable, shared by heads	0.458±0.011
Fixed, mild spread (0.9–1.1)	0.547±0.038
Fixed, wide spread (0.5–2.0)	0.716±0.064
ETTm1, steps 1–720, 3 seeds × 2 bases	
Fixed RoPE (scale 1)	0.767±0.051
Learnable, per head	0.488±0.033
Fixed 1.047 (= 1 day)	0.459±0.013
Fixed 0.955 (misaligned)	0.923±0.065
Fixed 1.10 (misaligned)	0.809±0.044

- **Head diversity**: for extrapolation, one shared scale performs as well as one per head; hand-set spreads help little or hurt. ⇒ no evidence the gain needs head specialisation.
- **Grouping**: the fast group learns the same value as the per-head scale; the slow group drifts with no measurable effect. ⇒ the effect sits in the **fast** frequencies.
- **Gradient clipping**: inactive after warm-up, so not the cause.

Backup: Transformer Basics

Architecture. Standard encoder–decoder with positional encodings (Vaswani et al., 2017).

Tokenization. History as tokens $\{x_1, \ldots, x_L\}$ (points or patches).

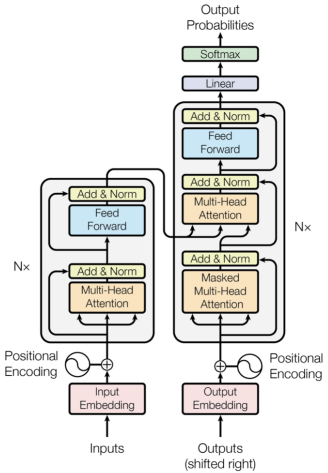
Projections.

$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (\in \mathbb{R}^{L \times d})$

Scaled dot-product attention (key equation)

$$\mathrm{Attn}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V$$

Multi-head attention allows different heads to capture diverse temporal patterns. Stacking layers builds contextual representations for sequence prediction.



Backup: Positional Encoding

Why do we need position encodings?

Self-attention treats the input as an unordered set of tokens — without additional signals, it cannot tell which time step comes first.

- To restore temporal order, each token x_ℓ is added with a **positional encoding**:

$$\tilde{x}_\ell = x_\ell + PE(\ell).$$

- The encoding $PE(\ell)$ can be:
  - **Sinusoidal:** fixed periodic functions $\sin(\omega\ell), \cos(\omega\ell)$ (Vaswani et al., 2017).
  - **Learnable:** parameters trained with the model (BERT-style).
- This addition injects absolute position information into the input before attention.

RoPE's idea

RoPE **replaces this absolute positional addition step** with a **rotation-based operation** inside the attention itself — making the attention score aware of relative offsets instead of absolute indices.

Backup: Why Relative Positions

- In NLP, **absolute positions** work because word order carries meaning.
- In time-series forecasting, especially in **financial data**, we care more about **relative offsets**:
  - Forecasting at $t{+}1$ should be invariant to where the window starts — the market pattern "5 days before a crash" should look the same in 2018 or 2024.
  - Temporal similarity depends on **distance** Δ, not on the absolute calendar index ℓ.
  - Financial series are often non-stationary — absolute time (e.g., day 2000 vs. day 4000) has little predictive meaning.

Motivation for RoPE

$$\underbrace{\begin{aligned} q_m &= f_q(x_m, m), \\ k_n &= f_k(x_n, n) \end{aligned}}_{\text{absolute position encoding}} \quad \Longrightarrow \quad \underbrace{\langle f_q(x_m, m), f_k(x_n, n) \rangle = g(x_m, x_n, m-n)}_{\text{relative position encoding target}}$$

Backup: From Rotation to Relative Encoding

Setup. Let d_h be even, $m = d_h/2$. Split $q, k \in \mathbb{R}^{d_h}$ into 2D blocks:

$$q = [q^{(0)}; \ldots; q^{(m-1)}], \quad k = [k^{(0)}; \ldots; k^{(m-1)}], \quad q^{(i)}, k^{(i)} \in \mathbb{R}^2.$$

RoPE rotation.

$$R(\omega_i \ell) = \begin{bmatrix} \cos(\omega_i \ell) & -\sin(\omega_i \ell) \\ \sin(\omega_i \ell) & \cos(\omega_i \ell) \end{bmatrix}, \quad q'^{(i)} = R(\omega_i \ell_q) q^{(i)}, \; k'^{(i)} = R(\omega_i \ell_k) k^{(i)}.$$

Dot-product score:

$$S = \frac{(q')^\top k'}{\sqrt{d_h}} = \frac{1}{\sqrt{d_h}} \sum_{i=0}^{m-1} q'^{(i)\top} k'^{(i)}.$$

Goal

Show S can be rewritten as a function of the **relative offset** $\Delta = \ell_k - \ell_q$.

Backup: Trigonometric Foundation of RoPE

Rotation matrix: $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

Compute:

$$R(\alpha)^\top R(\beta) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}.$$

Multiplication result:

$$= \begin{bmatrix} \cos\alpha\cos\beta + \sin\alpha\sin\beta & -\cos\alpha\sin\beta + \sin\alpha\cos\beta \\ -\sin\alpha\cos\beta + \cos\alpha\sin\beta & \sin\alpha\sin\beta + \cos\alpha\cos\beta \end{bmatrix}.$$

Apply trig identities:

$$\begin{cases} \cos(\beta-\alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha, \\ \sin(\beta-\alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha, \end{cases}$$

Hence:

$$R(\alpha)^\top R(\beta) = \begin{bmatrix} \cos(\beta-\alpha) & -\sin(\beta-\alpha) \\ \sin(\beta-\alpha) & \cos(\beta-\alpha) \end{bmatrix} = R(\beta-\alpha).$$

Composing two rotations equals a single rotation by the **angle difference** $(\beta - \alpha)$ — the key property enabling RoPE's relative encoding.

Backup: Relative Dependence Emerges

Key trigonometric identity.

$$R(\alpha)^\top R(\beta) = R(\beta - \alpha) \quad \Longleftrightarrow \quad \begin{cases} \cos(\beta-\alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha, \\ \sin(\beta-\alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha. \end{cases}$$

Apply to RoPE rotations.

$$q'^{(i)\top} k'^{(i)} = (R(\omega_i \ell_q) q^{(i)})^\top (R(\omega_i \ell_k) k^{(i)}) = q^{(i)\top} R(\omega_i(\ell_k - \ell_q)) k^{(i)}.$$

Therefore

$$\boxed{S(\Delta) = \frac{1}{\sqrt{d_h}} \sum_{i=0}^{m-1} q^{(i)\top} R(\omega_i \Delta) k^{(i)}}, \qquad \Delta = \ell_k - \ell_q.$$

Here the trigonometric identity merges the query and key rotations into a single relative rotation $R(\omega_i\Delta)$, so each 2D block interacts only through the relative offset Δ.

Backup: replication & hyper-parameters

- Code: `github.com/ycsama0703/ft5011-ropeformer` + `experiments/` (commands in `experiments/README.md`).
- Data: Time-Series-Library (HuggingFace, CC BY 4.0); EPF (Zenodo 4624805); exchange rates (Time-Series-Library).
- $L = 336$, patch 16, $d = 64$, 4 heads, 2 layers, FFN 128, dropout 0.1, batch 256.
- AdamW, lr 10^{-3} (no weight decay on scales), 300 warm-up steps + cosine, clip 1.0, bf16.
- Budgets: 2000 steps (ETTh1, EPF, exchange), 4000 (ETTm1), 6000 (Weather), 8000 (Electricity).
- RoPE base $\in \{10^3, 10^4\}$ and best checkpoint chosen on validation; test used once.
- One run: 20 s – 3 min on one H100 (memory capped at 8%).